

ADEPU PRANAV

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Education

Geethanjali College of Engineering and Technology

Expected May 2027

Bachelor of Technology in Electronics & Communication Engineering

Hyderabad, Telangana

- **CGPA:** 8.1 / 10.0
- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Computer Networks, Database Management Systems (DBMS), Operating Systems, Signals & Systems, Microprocessors & Embedded Systems

Sri Chaitanya Junior Kalasala

2021 – 2023

Telangana State Board of Intermediate Education (MPC)

Hyderabad, Telangana

- **Score:** 901 / 1000

Gitanjali Devashray

2021

Central Board of Secondary Education (CBSE)

Hyderabad, Telangana

- **Percentage:** 87%

Experience

Indian Railways Institute of Signal Engineering & Telecommunications

May 2025 – June 2025

Engineering Intern

Hyderabad

- Analyzed signaling architectures across **3+ operational zones**, documenting safety-critical protocols for real-time monitoring of **100+ train routes**.
- Evaluated **5+ railway signalling systems**, identifying operational gaps and compiling findings into a structured report for senior engineers.

Projects

BitTorrent Client (In Progress) | C#, Networking, Socket Programming

- Building a peer-to-peer BitTorrent client with bencode file parsing, tracker HTTP/UDP peer discovery, and parallel socket I/O for piece-based downloading.
- Engineered chunk verification using SHA-1 hashing, ensuring **100% data integrity** across all peer nodes.

Azure Healthcare Chatbot | Python, FastAPI, Azure AI

- Developed an AI chatbot handling **200+ concurrent queries** via Azure Bot Services and FastAPI with **sub-500ms response times**.
- Implemented multilingual NLP across **4 languages** (English, Hindi, Tamil, Telugu) and achieved **~90% intent accuracy** over **15+ healthcare categories**.

AI Video Summarizer | Python, Azure AI, NLP, MoviePy

- Engineered a pipeline achieving **95%+ transcription accuracy** via Azure Speech-to-Text, compressing 10-minute videos into 1-page summaries and cutting review time by **~70%**.
- Supported **3+ video formats** (MP4, AVI, MKV) with automated PDF export of key points.

ML-Based Log Anomaly Detection | Python, Scikit-learn, NLP

- Developed ML pipeline with **92% precision** and **88% recall** processing **1M+ log entries**, cutting anomaly detection time from hours to **under 5 minutes**.
- Designed dashboards monitoring **10+ anomaly categories** with real-time alerting for operations teams.

Technical Skills

- **Programming:** Python, C#, C, Java
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Technologies:** Machine Learning, NLP, Computer Networking, REST APIs, Azure AI Services, Socket Programming
- **Tools:** Git, GitHub, Unity, Ollama, Azure Bot Services, MoviePy

Leadership and Activities

- **COO, Gomi Inspections** — led a 4-member team to build and pitch an IoT Garbage Monitoring System at NER Hackathon, reaching **top 15 of 80+ entries**.
- **Vice President, Game Smiths Club** — organized **6+ events** per semester, growing membership by **30%** over one academic year.

Certifications & Achievements

- **Certifications:** Microsoft Certified: Azure AI Fundamentals, Microsoft Certified: Azure Fundamentals, HackerRank C# Basic
- **Achievements:** 1st Prize VNR VJIET Hackathon (250+ teams), Runner-up Hacktoberfest 2024 CBIT (700+ teams), Top 15 of 80+ at NER Hackathon Bengaluru